SMART INDIA HACKATHON 2026

- **Problem Statement ID - SIH26056**
- **Problem Statement Title - AI-Based Airfare Monitoring, Dynamic Indexing & CPI Integration Platform**
- **Theme -** Smart Automation / Governance & Economy / Transportation
- **PS Category -** Software
- **Team ID - 205**
- **Team Name - NULL POINTERS**
- **Tagline -** ***"Precision Analytics for Sovereign Airfare & CPI Intelligence"***

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AERONEX

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❖ Proposed Solution

AeroNex is an AI-assisted sovereign airfare monitoring, dynamic index calculation, and CPI integration platform that combines automated multi-source scraping, Laspeyres index computation, econometric forecasting, and MoSPI-aligned policy intelligence into one unified operational workflow.

Data Ingestion
OTA & Airline APIs
DGCA Tariffs

→

Pre-processing
Cleaning & Outlier
Normalization

→

Feature Eng.
Route Densities
Booking Lead Time

→

Laspeyres Engine
MoSPI 2012=100
Weighted Sub-index

↓

AI Price Forecast
LSTM / XGBoost
7-30 Day Outlook

←

Surge & Anomaly
Price-Gouging Spike
Monopoly Alert

←

Regional Radar
3D Globe & 2D Orbit
Metro Corridor Map

FINAL MOSPI ASSESSMENT

✓ STABLE (95–115)	! ELEVATED (115–135)	▲ HIGH SURGE (>135)
Standard Inflation	Advisory Notice	Flag Investigation

Key Features :

- **Multi-Source Extraction:** Automated live scrapers across IndiGo, Air India, SpiceJet, Akasa & leading OTAs.
- **Dynamic Laspeyres Engine:** Real-time airfare index calculated against MoSPI Base 2012=100 with regional weighting.
- **AI Price Trajectory:** Machine learning forecasts 7 to 30-day ticket swings with confidence boundaries.
- **Regional Route Radar:** 3D WebGL Globe & 2D SVG Orbit tracking 15+ major high-density metro corridors.
- **Surge & Gouging Flagging:** Proactive anomaly triggers during festival surges and emergency disruptions.

WHY AERONEX?

Existing approach	AERONEX
Manual Field Inspection Periodic sampled surveys by MoSPI staff with 30-45 day reporting delay.	**Multiple Real-time Signals** Automated 24/7 ingestion across 15+ hubs covering 1,000+ daily flights.
Static Spreadsheets Disconnected silos without route-level granularity or dynamic weights.	**Dynamic Laspeyres CPI** Sub-second index calculation mapped to MoSPI 3.02% transport group weight.
Lagged Review & Correlation Manual connection of disparate data weeks after price spikes occur.	**Explainable Risk Score** Evidence → Officer Review → Transparent Decision Trail.

TECHNICAL APPROACH

Frontend (User Interface)

- **React + TypeScript + Vite** — sub-second rendering, type-safe interactive intelligence dashboard.
- **Tailwind CSS + Recharts** — sovereign aviation theme (Dark Navy & Light), telemetry charts & gauges.
- **Three.js 3D Aviation Globe** — GPU-accelerated interactive flight arcs with automatic low-power device tiering.

Backend / Processing

- **Node.js + Express REST API** — high-throughput async processing and API gateway.
- **LiveDataStore & Ingestion Workers** — continuous 30s background price collection and index calculation.
- **Econometric & ML Pipeline** — Laspeyres index formulation, Z-score outlier filtering, and LSTM/XGBoost fare trajectory forecasting.

Data Layer

- **Supabase PostgreSQL** — persistent high-integrity store for historical fares, city pairs, and survey logs.
- **Row Level Security (RLS) & JWT** — authenticated role-based access for MoSPI field officers and analysts.
- **Real-time Data Streaming** — WebSocket channels for instantaneous push alerts on route price anomalies.

AIRFARE MONITORING & INDEXING FLOW

Live Scraping → Normalisation → Laspeyres Index → AI Forecast → Surge Alert

KEY OUTPUT SHOWCASE

140.2
/ 100 Base

▲ HIGH SURGE (+40.2%)

DETECTED SIGNALS

- Metro Peak: DEL-BOM (+39.3%)
- Fuel ATF Surcharge Index (+12.4%)
- Holiday Surge Anomaly Triggered
- Regional Load Factor: 88.5%

Automated MoSPI Notice & DGCA Tariff Alert

Audit ID: #IN-2026-904

FEASIBILITY AND VIABILITY

FEASIBILITY

Practical • Secure • Scalable • Impactful

1. TECHNICAL
Open-source TypeScript/Node stack, modular architecture, zero expensive third-party APIs.

2. OPERATIONAL
Automated scheduled scraping eliminates manual fieldwork overhead for MoSPI survey teams.

3. ECONOMIC
Low-cost serverless architecture, negligible ongoing compute costs, high ROI for governance.

4. SECURITY
TLS 1.3 encryption, Supabase RLS, role-based access control, and immutable audit logs.

5. TESTING
Extensive synthetic fare backtesting, Laspeyres unit testing, and cross-airline consistency tests.

6. DEPLOYMENT
Cloud/Edge ready; deployable on NIC Cloud, state servers, or sovereign government intranets.

RISK ASSESSMENT AND MITIGATION

Problem: Airline Anti-Scraping / Limits	→	Solution: Residential proxy pools & headless rotation
Problem: Extreme Volatility & Outliers	→	Solution: Rolling median smoothing & Z-score filters
Problem: Missing / Discontinued Routes	→	Solution: Dynamic route re-weighting algorithm
Problem: MoSPI Base Year Revisions	→	Solution: Configurable dynamic base year weights
Problem: Schema Drift & Payload Changes	→	Solution: Adaptive schema validators & Zod parsers

SCALABILITY

Built to Grow, Adapt & Scale Across India

1. MODULAR
Independent microservices.

2. 100+ AIRPORTS
UDAN regional hub growth.

3. HIGH VOLUME
100k+ daily fare quotes.

4. MULTI-CABIN
Business & air freight.

5. CLOUD / EDGE
NIC cloud & state centers.

6. API PORTAL
Zero-friction CPI ingest.

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IMPACT AND BENEFITS

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IMPACTS

1. MoSPI OFFICERS
Zero survey lag, live index ingestion.

2. DGCA REGULATORS
Rule 135 predatory pricing oversight.

3. AIR TRAVELLERS
Fair price alerts & AI fare forecasts.

4. AIRLINE CARRIERS
Transparent market benchmark indices.

5. POLICY MAKERS
Data-driven UDAN subsidy planning.

6. ECONOMISTS
Sovereign high-frequency datasets.

7. DECISION SUPPORT
Transparent inflation driver breakdown.

8. DIGITAL AUDIT
Traceable, verifiable fare records.

POTENTIAL BENEFITS

1. ECONOMIC ACCURACY
Eliminates distorted transport CPI inflation readings.

2. OPERATIONAL SAVINGS
90%+ reduction in physical airport survey costs.

3. MARKET TRANSPARENCY
Clear public oversight on airline dynamic pricing.

4. REGULATORY ENFORCEMENT
Automated detection of seasonal predatory spikes.

5. PREDICTIVE INSIGHTS
Forward-looking fare inflation forecasting.

6. NATIONAL SCALABILITY
Extensible to railway and freight logistics baskets.

EXPECTED OUTCOME

1. **Real-Time Airfare Index**
Sovereign, mathematically weighted Laspeyres index refreshed continuously 24/7.

2. **Direct MoSPI CPI Integration**
Automated mathematical contribution feed into the national 3.02% transport CPI basket.

3. **Proactive Surge Detection**
Immediate warning to DGCA Tariff Cell before price surges impact consumer welfare.

4. **Verified Sovereign Audit Trail**
Immutable historical fare logs ensuring complete integrity for public policy decisions.

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RESEARCH AND REFERENCES

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1. RESEARCH PAPERS

- **Laspeyres vs. Paasche Indices in Transport Pricing** — International Monetary Fund (IMF) Statistics Manual & Consumer Price Index Theory.
- **Machine Learning for Dynamic Airfare Prediction** — IEEE Transactions on Intelligent Transportation Systems & Applied Aviation Analytics (2022).
- **High-Frequency Big Data for CPI Measurement** — National Bureau of Economic Research (NBER) Working Papers on Scraped Pricing Data.
- **Econometric Analysis of Airline Dynamic Pricing & Monopoly Power** — Journal of Air Transport Management (2023).

2. DATA SOURCES

- **MoSPI CPI Datasets (Base 2012=100)** — Official transport and communication item baskets and rural/urban weights.
- **DGCA Domestic Aviation Statistics** — Monthly passenger carriage, route city-pair volumes, and airline market share data.
- **Carrier Portals & OTA Feeds** — Live flight schedules and fare quotes from IndiGo, Air India, SpiceJet, Akasa Air, and MakeMyTrip.
- **AeroNex In-Memory LiveDataStore** — High-frequency continuous polling engine logging real-time route fare quotes.

3. PATENTS & REGULATORY STANDARDS

- **Aircraft Rules, 1937 (Rule 135)** — Mandatory tariff display, predatory fare prevention, and DGCA Tariff Monitoring Cell regulations.
- **UN System of National Accounts (SNA 2008)** — Harmonized global standards for macroeconomic price indices and CPI compilation.
- **ICAO Doc 9562 & IATA Guidelines** — Economic regulation of air navigation services and passenger tariff transparent reporting.
- **ISO/IEC 27001 & CERT-In Guidelines** — Data protection, system integrity, and encryption standards for sovereign portals.

4. TECHNICAL DOCUMENTS & RESEARCH GAP

- **MoSPI CPI Methodological Manual** — CSO guidelines for price collection, imputation of missing quotes, and item basket weights.
- **DGCA Tariff Monitoring Cell Reports** — Historical market surge data during festive periods and natural emergencies.
- **Research Gap Addressed** — Bridging the critical lag between dynamic, algorithmic airline pricing and monthly static CPI surveys through sovereign AI intelligence.